Intelligence

Lecture 9

Giftedness & Child Prodigies

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Learning Outcomes

- Define ‘*giftedness*’, ‘*child prodigy*’
- Describe and evaluate key longitudinal studies of giftedness and prodigy
- Integrate cognitive, motivational, and contextual factors in theoretical explanations of prodigious talent.

Glossary of Key Terms

Giftedness
Exceptionally high ability or potential relative to age peers, often identified statistically (e.g. IQ $\approx$ 130+), but increasingly defined using multiple attributes (intellectual ability, creativity, achievement, motivation, leadership).

Child prodigy
A child (typically under age 10) who performs at an adult professional level in a demanding, culturally recognised domain (e.g. music, chess, maths), defined by realised achievement rather than potential.

Precocity
Rapid early mastery of knowledge or skills within a domain; often preferred to prodigy because it avoids mythic or "unnatural" connotations.

IQ (Intelligence Quotient)
A standardised measure of general cognitive ability; prodigies typically have above-average IQ, but extreme IQ is neither necessary nor sufficient.

Domain specificity
The principle that exceptional performance is usually confined to one domain (e.g. music, chess), rather than reflecting global superiority.

Glossary of Key Terms

Above-level testing
Assessment using tests designed for older individuals (e.g. SAT taken in early adolescence) to differentiate ability within the gifted range (used extensively in SMPY).

Working memory
The capacity to hold and manipulate information over short periods; consistently identified as a key strength in prodigies across multiple domains.

Rage to learn
An intense, sustained drive to acquire knowledge or mastery; a motivational feature often highlighted in extreme cases of prodigious achievement.

Co-incidence framework (Feldman)
A theoretical model proposing that prodigies emerge from a rare alignment of **natural talent**, **family and teacher support**, **domain readiness**, and **historical–cultural context**.

Sensitive periods
Developmental windows of heightened receptivity to learning; in prodigies, these may become unusually intense and individualised.

Gifted? A Prodigy?

Marie Curie

Serena Williams

Mozart

Magnus Carlsen

Giftedness

*The state of possessing a **great amount of natural ability, talent, or intelligence**, which usually becomes evident at a very young age. Giftedness in intelligence is often categorized as an IQ of two standard deviations above the mean or higher (**130 for most IQ tests**), obtained on an individually administered IQ test. Many schools and service organizations now use a combination of attributes as the basis for assessing giftedness, including one or more of the following: **high intellectual capacity, academic achievement, demonstrable real-world achievement, creativity, task commitment, proven talent, leadership skills, and physical or athletic prowess**. The combination of several attributes, or the prominence of one primary attribute, may be regarded as a threshold for the identification of giftedness.*

APA Dictionary of Psychology (2018)

Child prodigy

A prodigy is defined as a child who, at a very young age (typically younger than ten years old), performs at an adult professional level in a highly demanding, culturally recognized field of endeavor. A prodigy's performance is assessed as being at a professional level based on the standards of their field as well as the reaction of the general audience, reflected, for example, in sales of paintings and positive reviews of performances.

Feldman with Morelock (1986)

Gifted vs Prodigy

Gifted

- Refers to exceptionally high ability or potential relative to age peers
- Often defined statistically (e.g. top 2–10%, IQ ≈ 130+)
- Ability may be general (high reasoning / intelligence) or multi-domain
- Emphasis is on capacity, not necessarily outstanding achievements
- Can be identified at different ages and often persists across the lifespan

(Gagné, 2004, 2010; APA Dictionary of Psychology; Morelock, 1996)

Prodigy

- A child who achieves adult-expert or professional-level performance
- Almost always domain-specific (music, chess, maths, art)
- Defined by realised achievement, not just potential
- Emerges very early, often before age ~10
- The label usually fades in adulthood, even if high ability remains

All prodigies are gifted, but most gifted children are not prodigies (extremely rare even among gifted children)

(Feldman, 1986; Ruthsatz & Urbach, 2012; Shavinina, 1999)

Giftedness

Terman's Termites

Giftedness – *Terman's Termites*

Lewis Terman

GENIUS AND STUPIDITY

A Study of some of the Intellectual Processes of Seven "Bright" and Seven "Stupid" Boys

Terman's PhD work

- Terman studied intelligence in a group of children from an early age into adulthood
- This group of children became known as *Terman's Termites*

Giftedness – *Terman's Termites*

Lewis Terman

- Using Stanford-Binet and other tools, Terman's team scoured California schools to **identify high IQ children**
- He argued that **special attention should be paid to those with high scores** (identified early and be accelerated through school)
- Identified a core group of 643 children with **IQs of 135** or higher
- By 1928, Terman had 1,528 subjects between the ages of 3 and 28.
- Amassed a thick dossier detailing their lives and **followed how their lives and careers developed**

Giftedness – *Terman's Termites*

Lewis Terman

- Kids proved remarkable in some ways and ordinary in others
- One distinction was a **keen pursuit of higher studies** (2/3rds earned bachelor's degrees – 10x the national rate for their time)
- They also **flocked to grad school** – there were 97 PhDs, 57MDs, 92 lawyers.
- The **women had fewer children and focussed more on higher education and careers**
- In other ways, the Terman kids were 'normal' – they got divorced, committed suicide and became alcoholics at about the national rate.

Giftedness – *Terman's Termites*

Lewis Terman

- In later work, Terman suggested that individuals scoring two standard deviations above the sample mean IQ were, on average, superior across physical, moral, and behavioural domains (Terman & Oden, 1947; 1959).
- Terman **advocated the widespread use of intelligence tests** in childhood as a way to identify highly able individuals and to **nurture their talents through tailored education**.
- However, these conclusions were drawn within the early-20th-century eugenics framework that influenced Terman's thinking, **which we must critically confront - now viewed as historically and ethically problematic!**

Giftedness – *Terman's Termites*

Lewis Terman

Limitations of Terman's Study:

- **Sampling bias**: Participants were not randomly selected and were overwhelmingly white, middle-class, and Californian.
- **Lack of a comparison group**: No matched control sample, limiting causal conclusions.
- **Researcher interference**: Terman intervened in participants' lives, potentially shaping outcomes.
- **Historical specificity**: Findings reflect a particular social and historical context (Depression, WWII), limiting generalisability.
- **Identification bias**: Gender imbalance and severe under-representation of minority groups raise questions about who was labelled "gifted."

Giftedness

Study of Mathematically Precocious Youth (SMPY)

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

- Began in 1971 (Julian Stanley)
- One of the **longest-running longitudinal studies** in psychology
- Tracks **>5,000 intellectually precocious individuals from adolescence into adulthood**
- **Focus**: *How early cognitive ability relates to education, career paths, and real-world achievement*

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

- Mostly identified at ages 12–13

- Selected using above-level testing:
  >Took the SAT 4-5 years earlier than typical college-bound students

- Cohorts ranged from:
  >Top 1%
  >Top 0.5%
  >Top 0.01% in mathematical or verbal reasoning

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

Why above-level testing matters?

- Normal school tests have **ceilings** → gifted students all score near the top

- Above-level tests:
  - Reveal **meaningful differences within the gifted range**
  - Distinguish “very able” from “exceptionally able”

- These early differences **predict long-term outcomes**

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

Key question of the paper:

Is there an ability "ceiling"?

Does ability stop mattering once people are already very bright?

SMPY answer: No.

Differences **within the top 1%** still strongly predict adult outcomes

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

Main findings: careers & impact

Compared with population averages, SMPY participants were far more likely to:

- Secure **tenure-track university positions**
- Hold **high-income jobs**
- Earn **patents** (indicator of innovation)
- Work in **STEM leadership roles**

- Early SAT scores predicted:
- Patents
- Publications
- Faculty positions at top universities

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

Even *within* the top 1%:

Higher early SAT scores → greater likelihood of:

- Doctorates
- STEM careers
- Tenure at elite institutions

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

Interests & motivation matter too

The study showed that ability alone is not enough:

Career paths depend on:
- Interests
- Values
- Motivation

- Gifted people often choose different fields by preference, not because of lack of ability

- Women and men showed:
  >Similar levels of success
  >Different field distributions, partly reflecting interest patterns

Giftedness – *Study of Mathematically Precocious Youth (SMPY)*

Big picture takeaway

Early cognitive differences matter - even at very high levels

But:

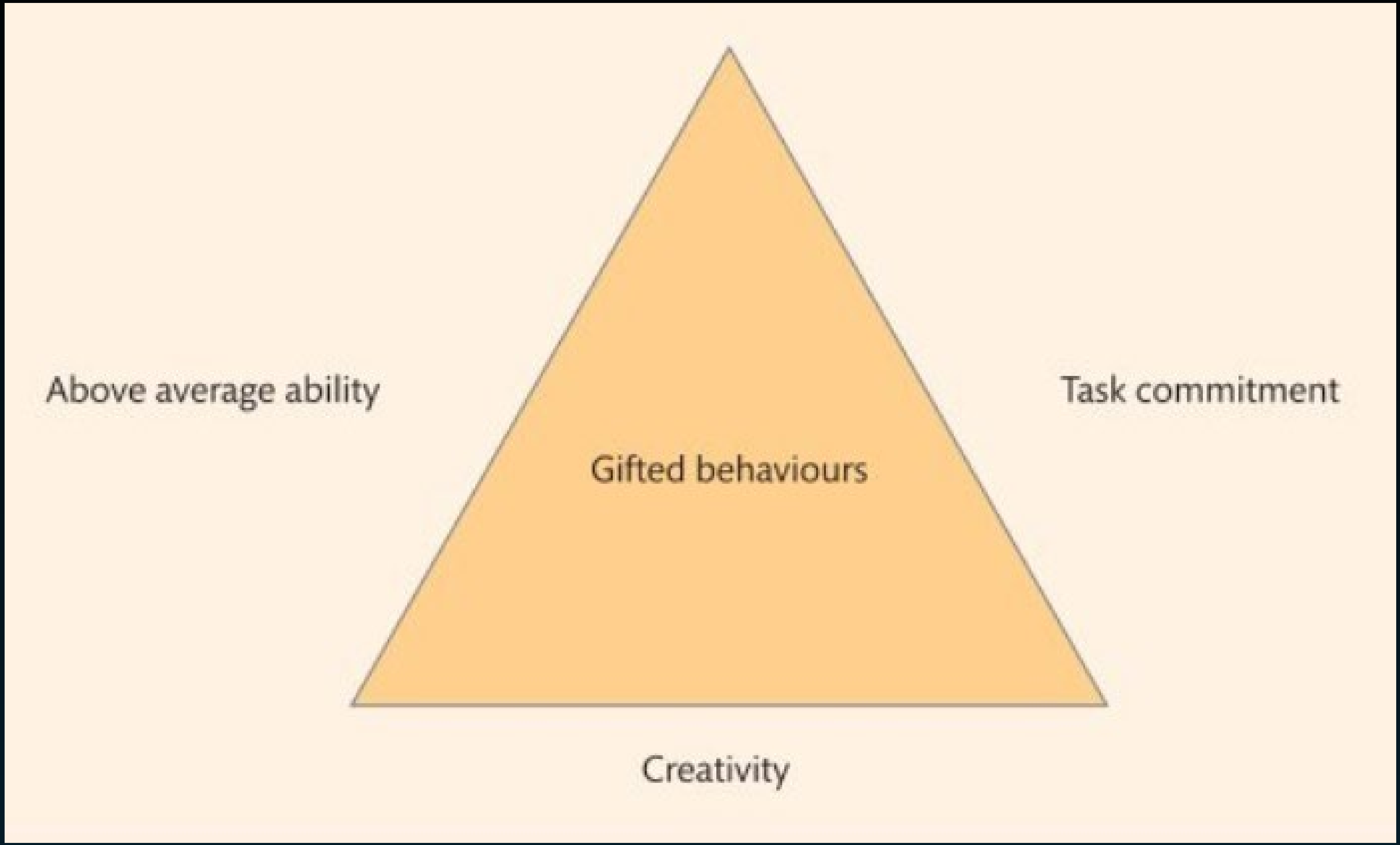
- Ability ≠ destiny
- Outcomes reflect:
    - Ability
    - Opportunity
    - Educational fit
    - Motivation and interests

*Talent development is a **long-term process**, not a single test score.*

What factors typically indicate Giftedness? - Council for Exceptional Children, 1990

- shows superior reasoning powers and marked ability to handle ideas; can generalise readily from specific facts and can see subtle relationships; has outstanding problem-solving ability
- reads avidly and absorbs books well beyond their years
- sustains concentration for lengthy periods and shows outstanding responsibility and independence in classroom work
- shows insight into arithmetical problems that require careful reasoning and grasps mathematical concepts readily
- learns quickly and easily and retains what is learned; recalls important details, concepts and principles
- has a wide range of interests, often of an intellectual kind; develops one or more interests to considerable depth
- shows social poise and an ability to communicate with adults in a mature way
- gets excitement and pleasure from intellectual challenge
- shows persistent intellectual curiosity; asks searching questions; shows exceptional interest in the nature of man and the universe
- comprehends readily
- shows an alert and subtle sense of humour
- shows initiative and originality in intellectual work; shows flexibility in thinking and considers problems from a number of viewpoints
- sets realistically high standards for self; is self-critical in evaluating and correcting his or her own efforts
- shows creative ability or imaginative expression in such things as music, art, dance, drama; shows sensitivity and finesse in rhythm, movement and bodily control
- is markedly superior in quality and quantity of written and/or spoken vocabulary; is interested in the subtleties of words and their uses
- observes keenly and is responsive to new ideas

Renzulli's three-ring definition of giftedness



Child prodigies

Feldman & Goldsmith, 1968

Child Prodigies

- Contemporary research on child prodigies can be traced to the publication of a seminal study of six boys under the age of ten (Feldman & Goldsmith, 1968).
- The boys were **between three and eight years of age** when first studied in the fields of **music, chess, and writing** (and an “omnibus prodigy,” who had not yet settled into a specific area)
- Participants were **followed longitudinally for up to ten years.**
- Examined both **specific domain abilities** and **general cognitive abilities.**
- It explored the **boys’ experiences with teachers and family members**.
- Development in each child’s specific talent domain was considered within the context of their broader overall development.

Feldman & Goldsmith, 1968

Child Prodigies

- Data collected through observation in natural settings, recordings of practice, and interviews with the children, parents and teachers.

- Focus on both **specific talent** (e.g. composition, problem-solving) and **general abilities** (IQ, school achievement).

- Tracked experiences with schooling, family support, and access to expert instruction.

- Interpreted within a developmental framework (how the specific talent unfolded alongside broader cognitive, social and emotional development).

Feldman & Goldsmith, 1968

Child Prodigies

Key Findings

- Child prodigies show a **blend of child-like and adult-like characteristics**.
- Extreme performance depends on **sustained support from at least one parent plus teachers/mentors**.
- Even in the most extreme cases, **reaching adult-level expertise takes several years of intense engagement**.
- They argued that talents are at least partly natural/inborn; the **more extreme the case, the more strongly inborn factors appear to contribute.**
- Prodigies' talents are **highly domain-specific** (e.g. music, maths, chess) rather than globally superior across all areas.
- General intelligence is **typically above average** but not necessarily at an extreme "genius" level.

Feldman & Goldsmith, 1968

Child prodigies

Howard (2008)

Child Prodigies

Howard (2008) – Study of Eight Chess Prodigies

- Followed eight child chess prodigies, defined as children who had already reached professional tournament level.

- Asked whether exceptional child performance predicts exceptional adult performance in the same domain (chess).

- Used this to **probe natural talent vs. practice in reaching world-class levels**.

- Responded to the claim that **few prodigies become successful adults in their original field**.

- In chess, **these child prodigies generally did become successful adult performers in chess**.

- Howard **argued for the importance of natural talent as a key ingredient** in chess success; the **prodigy phenomenon is hard to explain without a substantial natural talent base** (as argued by Feldman, 1995, 2008; Morelock & Feldman, 1993, 1999; Winner, 1996).

Howard, 2008

Child Prodigies

- Argued that most of the eight prodigies achieved high levels of international success as adults, despite having less total practice than many weaker players.

- On several indices, they outperformed other high-level chess players, for example:
  > Needed fewer games to reach master level
  > Took fewer years to become grandmasters
  > Were younger when they obtained grandmaster ratings

- One of the eight became a world champion, although not all world champions would count as child prodigies under the study’s specific definition.

- Interpreted as evidence that exceptional, possibly inborn talent must play a role, because these prodigies outperformed peers despite generally having accumulated less practice.

Howard, 2008

Limitations and Criticisms

Measurement of practice/study time is limited

- Howard estimated chess study using a small number of self-report questions in an online survey. Critics argue this substantially underestimates true practice, compared with the standard expert-performance method of 30-minute retrospective interviews tracking yearly engagement across multiple practice activities (see Ericsson & Moxley, 2012).

Confounding of performance and opportunity

- Howard treats number of international games/tournaments played as partly reflecting talent (i.e. “these prodigies play many high-level tournaments because they are exceptionally able”)
- **Tournament participation is not a pure measure of talent** because who gets to play more international tournaments often depends on resources and prior success, not just underlying ability.
- Participation in international tournaments is costly (airfare, accommodation, entry fees).
- Critics argue that higher ratings enable greater tournament participation, rather than tournament play reflecting innate talent independent of training.

Howard, 2008

Child prodigies

Ruthsatz & Detterman (2003)

Child Prodigies

- Used a detailed case study to examine the **role of general intelligence (IQ)** in a young piano prodigy.
- They argued that IQ played an important role in the child's ability to perform at **an adult, professional concert level** by the age of six.
- In addition to **strong domain-specific musical skills**, the child demonstrated **an IQ in the gifted range**, which contributed to his overall level of performance.
- A particularly striking finding was the child's **exceptional general memory** and music-specific memory abilities.
- The study **challenged the idea that practice alone** explains prodigious performance, noting that the child had **received little or no formal musical training** at that stage.

Ruthsatz & Detterman, 2003

Child Prodigies

- Overall, the findings suggest that high-level musical performance reflects a combination of:

  > elevated general intelligence,
  > domain-specific natural abilities, and
  > practice (even if limited early on).

- This combination-based explanation aligns with broader theoretical accounts of **prodigies and savant abilities.**

- Note, the study is based on **one individual**, which limits how far the findings can be generalised to other prodigies.

- The absence of extensive formal training does not rule out other forms of **informal practice or exposure** to music.

Ruthsatz & Detterman, 2003

Child prodigies

Ruthsatz & Urbach 2012

- Examined 8 child prodigies across different domains (music, art, maths, gastronomy).
- **Aim**: to characterise their cognitive profile and explore links with autistic traits and savant skills.
- IQ scores varied from above average to highly superior, indicating that **prodigies are not uniformly at extreme IQ levels**.
- **Exceptionally high working memory** was the most consistent cognitive feature across all prodigies.
- Prodigies showed **very high attention to detail**, a trait often associated with autistic cognitive styles.
- Elevated autistic traits were observed in several prodigies and in first- and second-degree relatives.
- The authors suggest that prodigies may share some cognitive characteristics with autistic savants, particularly strong memory and detail-focused processing.

Ruthsatz & Urbach, 2012

Child prodigies

Edmunds and Noel 2003

Child Prodigies - Geoffrey

- Edmunds & Noel (2003) reported a detailed **case study of an exceptionally advanced young writer**, focusing on a 12-month period from age 5 to 6.
- During this period, the child (Geoffrey) produced **129 separate works**, totalling **over 1,500 handwritten pages**.
- His writing often reflected interests in **maths and science**, though some early work drew on popular culture (e.g. Pokémon), written for a younger sibling.
- Writing was typically produced **very rapidly**, described as occurring in bursts of intense creative energy.
- On standard language measures, Geoffrey’s writing exceeded norms for high school students.
- General intelligence: WISC-III IQ score: **128** (moderate-to-high range). Raven’s Matrices: **above the 99th percentile** for a much older comparison age. Authors also noted **exceptional memory**, including detailed recall of earlier work.

Child Prodigies - Geoffrey

Dear Jim,

I am into math but also science. Here's the math part. I know addition, addition with tens and ones, multiplication, multiplication with tens and ones, division, and division by zero!! Here's how that works. 5 [divided by] 0 = undefined, or, the answer is undefined. I can do algebra, addition with tens, ones, hundreds, thousands, and millions up to infinity. . . . I also have a bunch of questions. What is calculus? . . . How do you get –0 if it exists?

Now, some science. I do theoretical physics just like you. I am working on a unified theory. Are you? And if you're not, what's the theory you're working on anyways? . . . My unified theory is broken up into many parts, each part the size of special relativity . . . E = sp, meaning energy = speed of light pulses. It is the theoretical answer to why Pikachuic electricity is so fast. . . . I really know my geometry, even though I'm in grade 1! I know that a rhombicosidodecahedron has 240 forces. A rhombicosidodecahedron is the largest known polyhedron. It is huge!

XOX
Geoffrey

Edmunds & Noel (2003)

Child Prodigies - Geoffrey

- **Motivational characteristics**: The most striking feature was a strong, persistent drive to learn, described as a “dogged persistence” or “rage to learn,” a trait often observed in the most extreme cases of early achievement.

- **Conceptual framing**: Edmunds & Noel preferred the term “precocity” rather than “prodigy”, emphasising rapid early mastery and avoiding definitional problems associated with prodigies.

- **Contribution to the literature**: Writing prodigies are extremely rare, and this case adds rich, domain-specific developmental detail to a very small research literature on prodigious achievement in writing.

Edmunds & Noel (2003)

Child prodigies

Comeau et al. 2017

Child Prodigies

- Context: Musical prodigies are **by far the most common type of prodigy**, making them a crucial test case for understanding extreme early talent.

- **Aim**: Comeau et al. (2017a, 2017b) examined whether musical prodigies genuinely perform at **adult professional levels**, and **which specific abilities** support this performance.

- Cognitive & musical profile (2017a): An 11-year-old piano prodigy (LN), compared with other musicians and a historical case (Nyireghazi), **showed adult-level performance, high (but not extreme) IQ**, **exceptionally strong working memory**, and strong **parental and teacher support**.

- **Uneven skill profile**: Prodigious musical talent reflected **domain-specific strengths** (e.g. pitch accuracy) alongside **average or weaker abilities** (e.g. improvisation), rather than global superiority.

- **Perceptual evidence**: In listening tasks, prodigies were often **indistinguishable from professional musicians**, particularly **older prodigies (11–14)**, with trained musicians only marginally better than non-musicians at telling them apart.

Child prodigies

Theoretical Interpretations

Child Prodigies - Theoretical Interpretations

Persistent definitional problems:
The term *prodigy* carries historical and cultural connotations of being “unnatural,” leading many scholars to prefer **precocity**, which emphasises rapid early mastery without invoking mystique.

Limits of IQ-based explanations:
Prodigies usually show **above-average intelligence**, but **extreme IQ is neither necessary nor sufficient**; exceptional domain-specific performance cannot be explained by general intelligence alone.

Domain specificity and structure:
Prodigies emerge primarily in **highly structured domains** (e.g. music, chess, maths), where progress and excellence can be clearly defined and measured.

Child Prodigies - Theoretical Interpretations

Feldman’s co-incidence framework:
Prodigy arises from a rare alignment of natural endowment, intensive family and teacher support, a developmentally receptive domain, and a supportive historical and cultural context.

Developmental and motivational factors:
Theories emphasise distinctive cognitive and emotional features, including intense, sustained motivation (“rage to learn”), beyond what IQ tests capture.

Sensitive periods reinterpreted:
Prodigies may experience individualised sensitive periods, where early interests become unusually deep and stable, underpinning long-term commitment and expertise.

Overall Summary

- Lecture distinguishes giftedness (high potential) from child prodigy (early professional-level achievement in a specific domain). While all prodigies are gifted, most gifted children are not prodigies.

- Evidence from major longitudinal studies (e.g. Terman, SMPY) shows that early cognitive ability predicts later educational and occupational outcomes, even among very high-ability individuals. However, ability alone does not determine life success; interests, motivation, and opportunity play crucial roles.

Overall Summary

- Research on child prodigies demonstrates that prodigious performance is domain-specific, develops over time, and does not require extreme IQ. Instead, prodigies typically show above-average intelligence, exceptionally strong working memory, intense motivation, and sustained support from families and expert teachers.

- Prodigies cannot be explained by a single factor such as IQ or practice. Instead, they emerge from a rare alignment of cognitive abilities, motivational characteristics, environmental support, and cultural–historical context.